



Hydrodomus

Small hydrogen refuellings stations at home

"EV wall charger for hydrogen"



LA
FRENCH TECH
GRAND PARIS



The problem



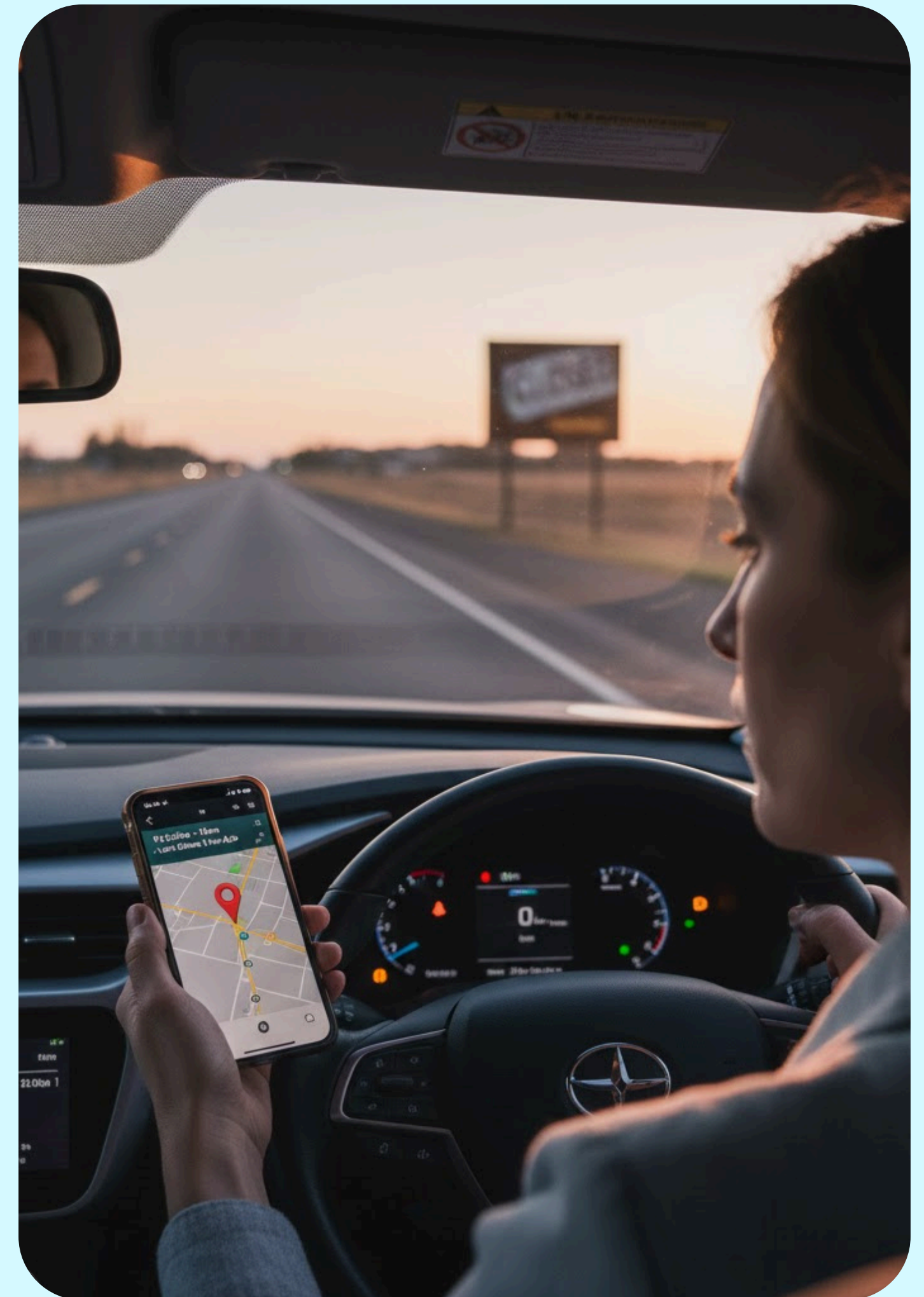
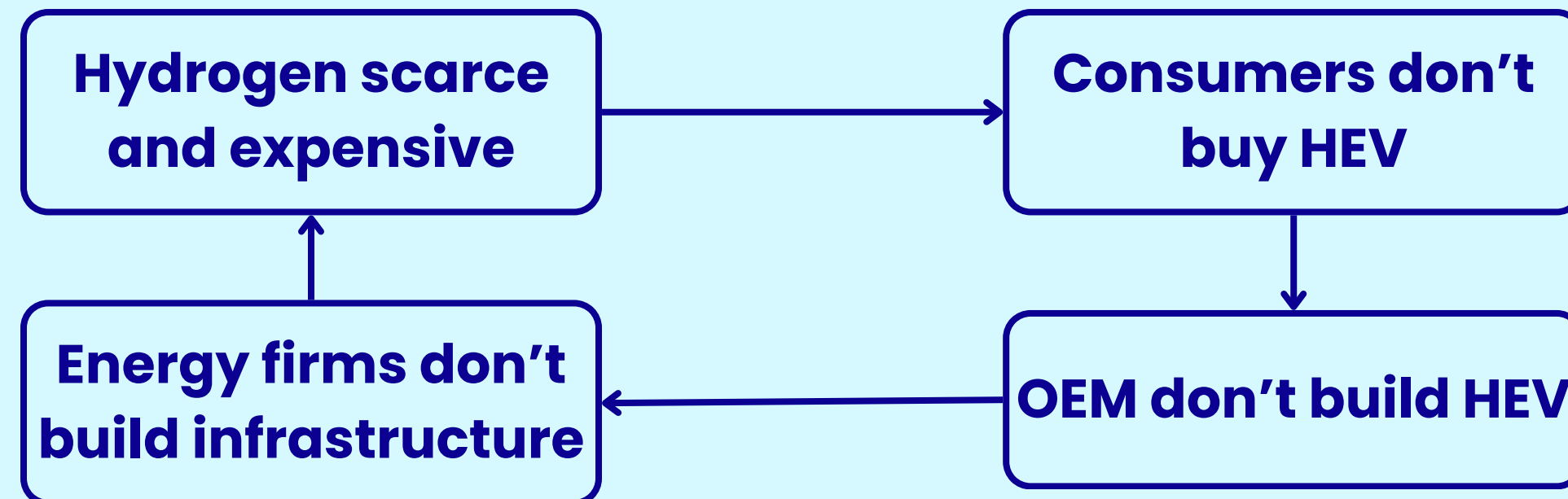
Poor **accessibility** (few stations) and high **cost**



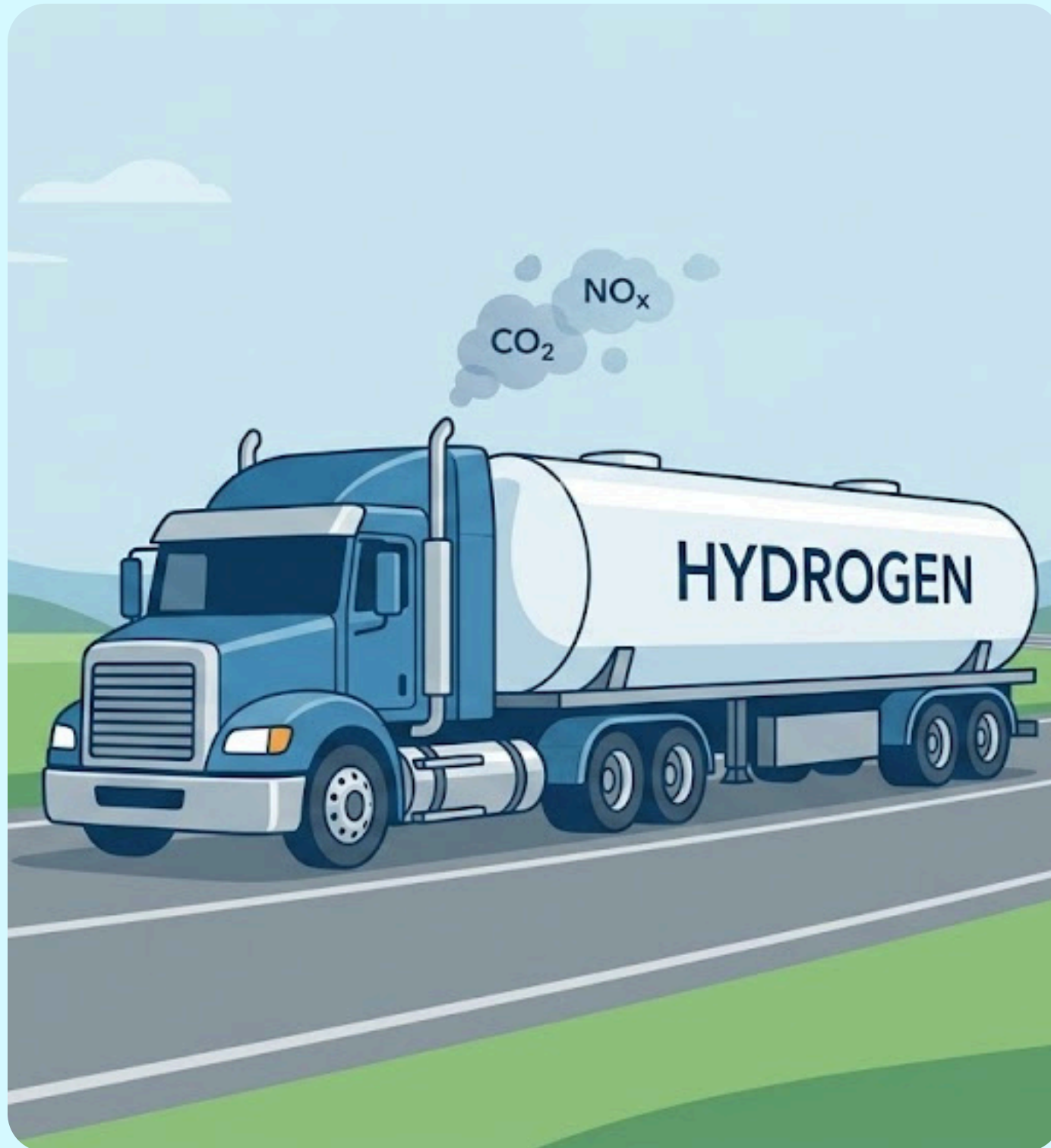
85% of hydrogen is not **green** in the stations



The hydrogen is expensive in the current stations



The problem



Hydrogen is expensive and not clean because the distribution is wrong

1. The hydrogen is produced in great quantities from non-green processes
2. It is distributed by pipes and trucks to the stations
3. Consumers have to drive there to refuel

This is the same model than fossil fuels :
centralisation of the production.

The solution



Hydrodomus concept :
"Generate your hydrogen at home"



Only uses water and electricity:
no green-house effect gases



The end-user is fully autonomous:
they own the supply.



High safety of the system,
following every regulations



How it works



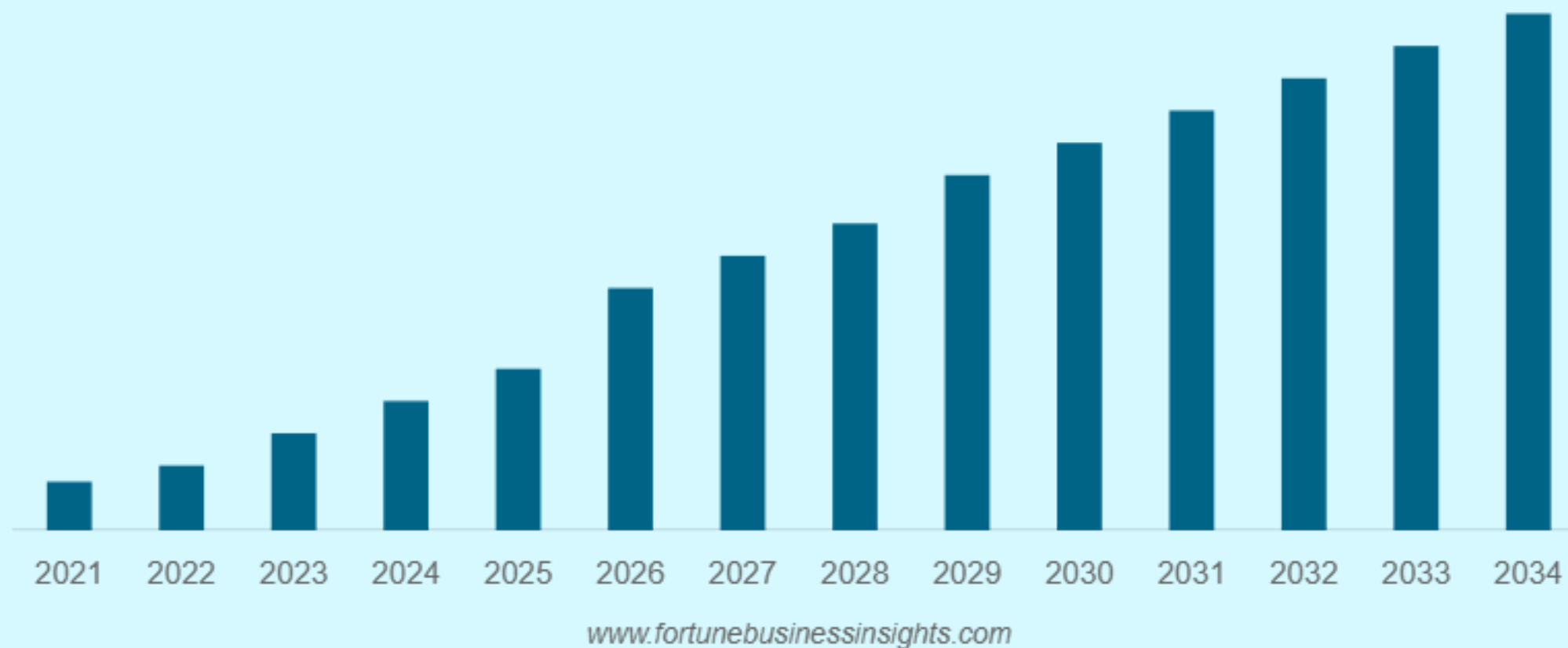
1. Generate Hydrogen gas by electrolysis
2. Compress the gas at 700 bars
3. Send it in the bottle inside the car



Market validation

Car manufacturers (~\$15B total spent on hydrogen cars development) :
Toyota, BMW, Hyundai, Honda, Mercedes, Alpine, Renault and NamX

Europe Fuel Cell Vehicle Market Size, 2021-2034 (USD Million)



Key insights:

- ~90,000 FCEV on the road today
- 2.5M FCEV on the road by 2030
- CAGR = 55%
- Japan: Mandating 800,000 FCEVs on the road by 2030.
- EU law now requires hydrogen refueling stations at least every 200 km

Price of H2 today : **15-30 \$/kg**

Price of H2 with Hydrodomus : **4-7 \$/kg**

Market Size

Global Fuel Cell Vehicle Market Share, By Vehicle Type, 2026



www.fortunebusinessinsights.com

Metric	Value	Reference
TAM	€170B+	Global Hydrogen Infrastructure & Home Energy Storage (2040 Horizon)
SAM	€15B	2.5M Passenger FCEVs projected by 2030 (Light Duty Vehicles).
SOM	€250M	High-utilization fleet/taxi drivers in EU/US (Immediate 3-year target).

Benchmark

	Public Stations (Air Liquide/HRS)	Toyota Cartridges (Prototype)	HydroDomus	Battery EV
Convenience	Low (Must travel to station)	Medium (Manual handling/swap)	High (Refuel while you sleep)	High (overnight charge, grid-dependent)
Infrastructure Cost	High (~€3M+ per station)	High (Logistics & swap hubs)	Medium (consumer pays himself)	Low for light use; grid upgrades needed for heavy-duty
Fuel Source	Often Grey/Industrial H ₂	Varies (logistics-dependent)	100% Green (Self-generated)	100% electricity
User Experience	"Gas Station" model	"Propane Tank" model	"Home refuelling" model	Simple UI; limited for heavy-duty or off-grid
Logistics	High-Intensity Industrial	Complex Circular Logistics	Zero Logistics. Hydrogen made on site	Grid infrastructure upgrade required for heavy-duty

The Tech: Efficient, Safe & Scalable

Numbers :

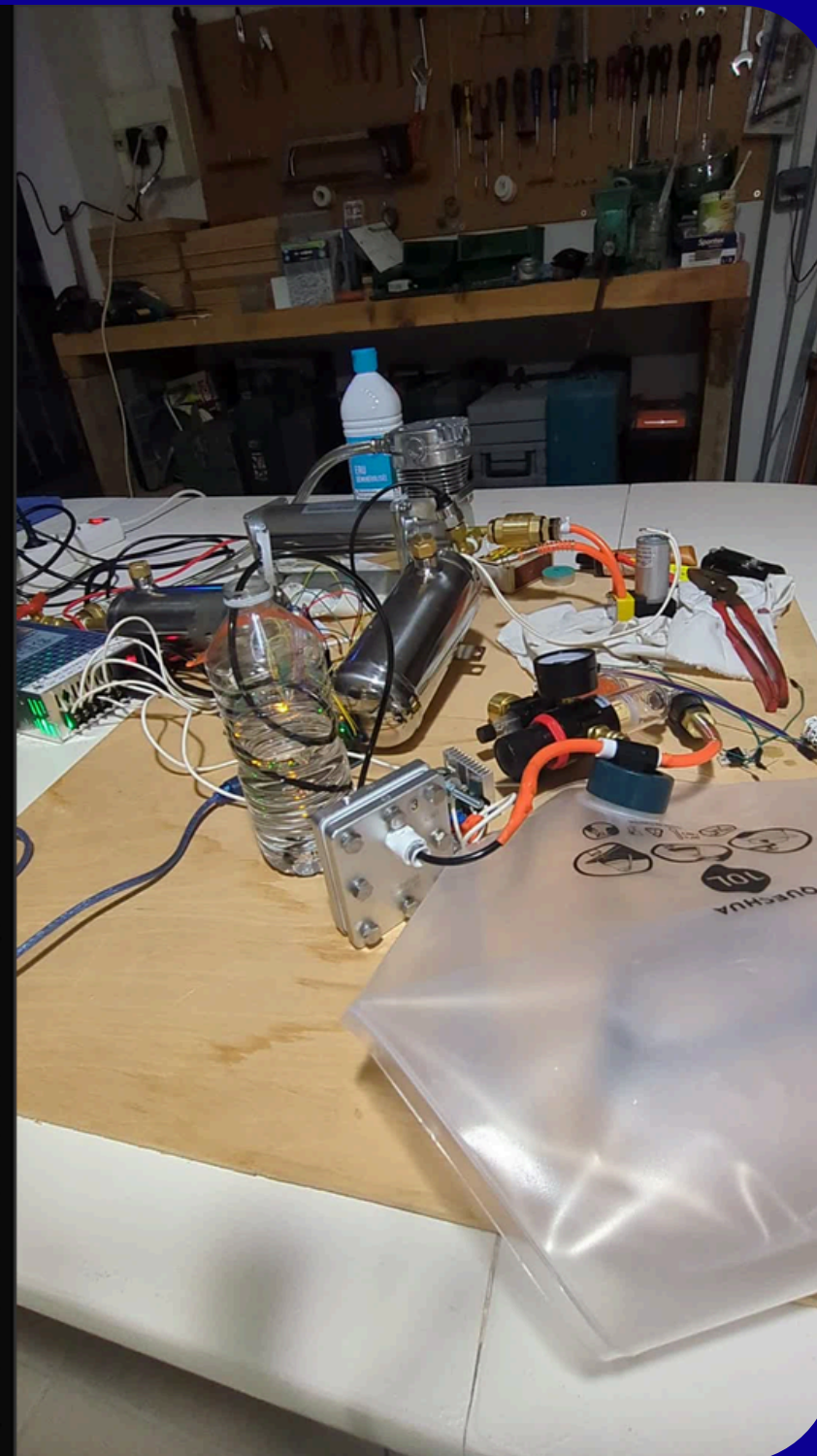
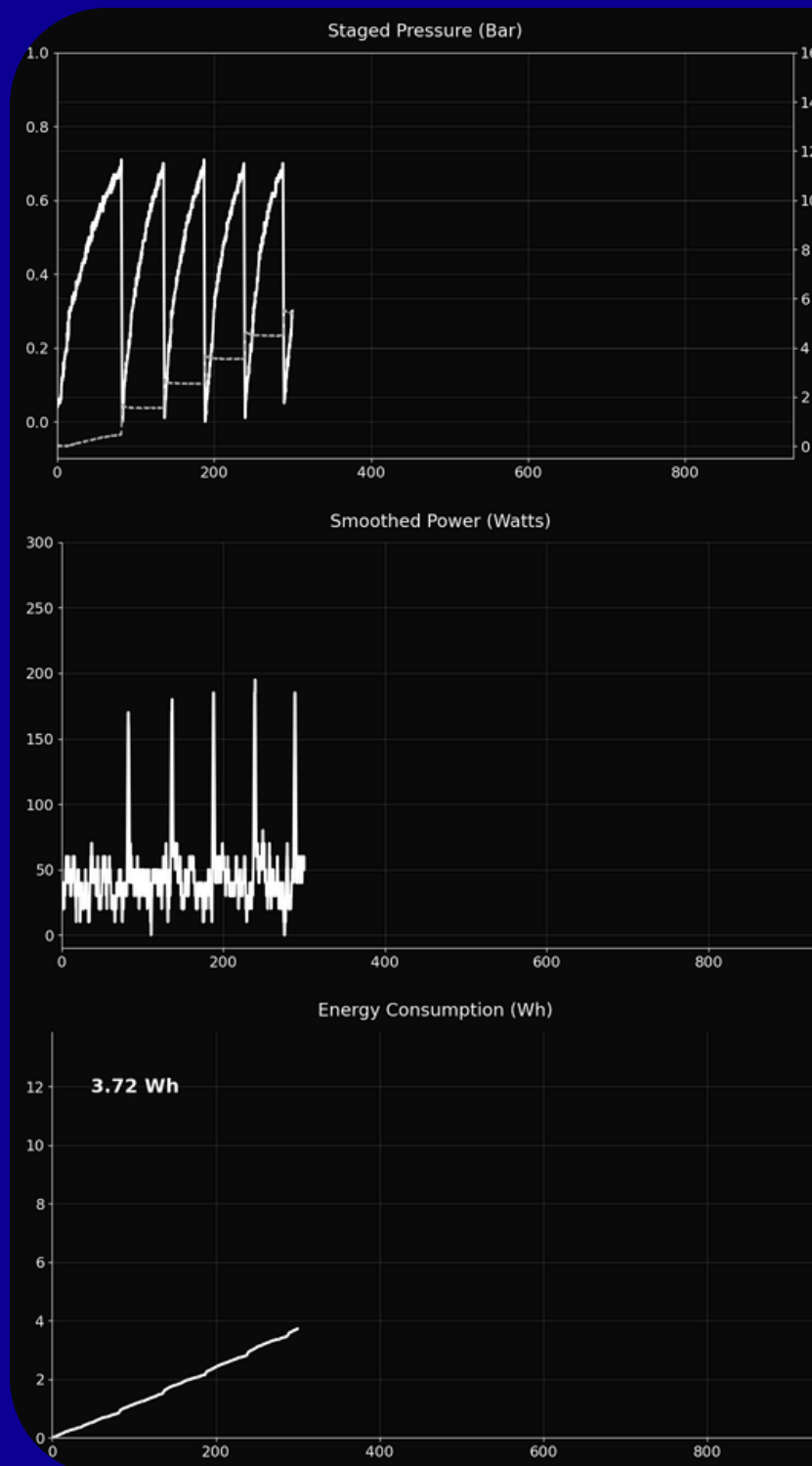
Efficiency: ~65%

Pressure: 700 bar

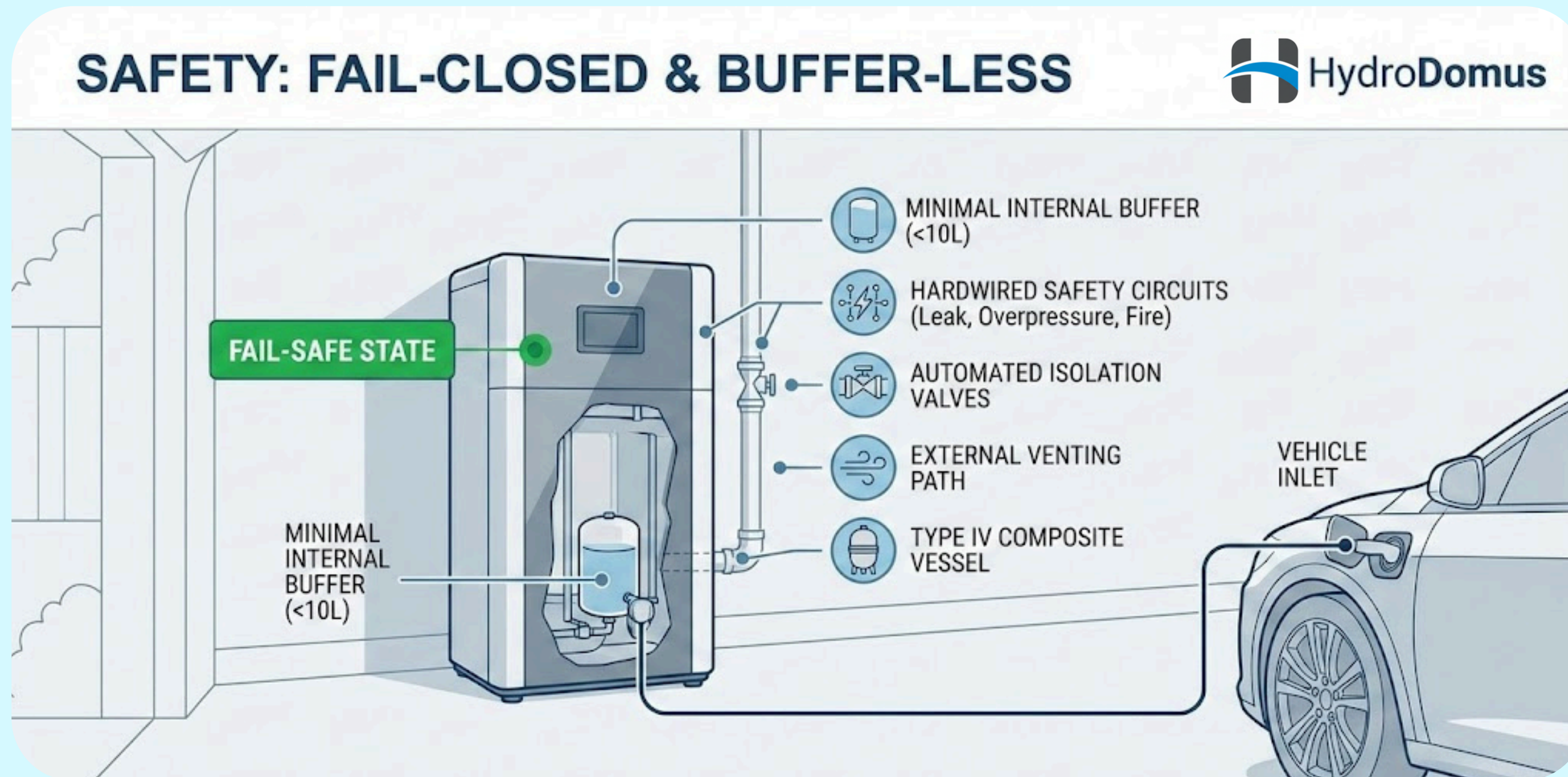
Purity: 99,99% (grade 4.0)

Refueling overnight

1kg of H₂ generated every 24h



Safety



Most of the norms and certifications will be met by the suppliers

Norms and certifications :

- Hydrogen fueling : **ISO 19880**
- Refueling interface : **ISO 17268 / SAE J2600 / SAE J2601**
- Explosive atmosphere : **ATEX 2014/34/EU**
- PEM : **ISO 22734**
- Storage : **Type III / IV**
- Pressure Equipment : **PED 2014/68/EU**
- Machinery : **2006/42/EC**
- Electricity : **2014/30/EU**

The prototype

- **Status:** Subscale bench prototype built and operational.
- **Tech Validated:** PEM generation (6 L/h), multi-stage filtration, and 2-stage mechanical compression (14 bar).
- **Strategy:** Subsystems decoupled and tested independently to ensure maximum safety and low-cost validation before full 700-bar integration.



The Founding team



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Business development & Engineering

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The advisors



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Hydrogen cars | Hyvia founder

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Connections to suppliers and
customers

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Common questions

- **Why choosing hydrogen cars over BEV ?**
 - No batteries; ~3 min refuel; longer range; better cold-weather performance; less grid strain than mass BEV charging.
- **Hydrogen cars are less efficient than BEV, why bother ?**
 - We're not competing with BEVs on efficiency, we provide refueling infrastructure for FCEVs that already exist on the road (Mirai, Nexo, iX5).
- **How can this system produce 700 bar of hydrogen and stay safe ?**
 - High-pressure buffer capped under 10 liters (vs. hundreds of kg at public stations), fail-closed valves isolating the system on fault/power loss, plus forced ventilation and leak-before-burst tank design.